

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	29 June 2026
Project Name	Network Request Management System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

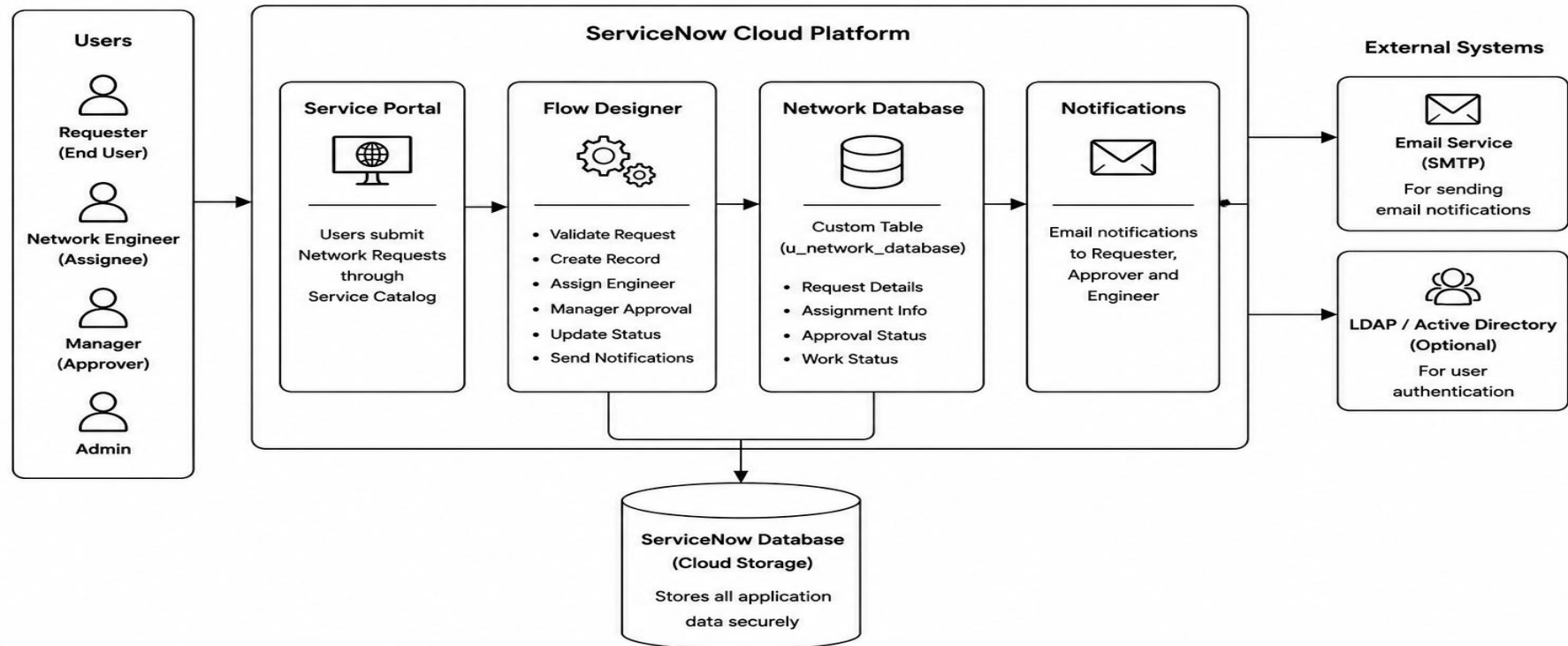


Table-1 : Components & Technologies:

S.No	Component	Description	Technology Used
1	User Interface	Allows users to submit and track network requests through a service catalog.	ServiceNow Service Portal, Service Catalog
2	Application Logic-1	Validates request details and creates a network request record.	ServiceNow Flow Designer
3	Application Logic-2	Assigns the request to the appropriate network engineer and routes it for manager approval.	Flow Designer, Approval Engine
4	Application Logic-3	Sends email notifications to the requester and approver based on request status.	ServiceNow Notification Engine
5	Database	Stores network request details, assignment information, approval status, and work status.	ServiceNow Custom Table (u_network_database)
6	Cloud Database	Provides secure cloud-based storage for application data.	ServiceNow Cloud Database
7	File Storage	Stores attachments related to network requests (documents, screenshots, etc.).	ServiceNow Attachment Service
8	External API-1	Sends email notifications for request creation, approval, and completion.	SMTP / ServiceNow Email Service
9	External API-2	Integrates user and manager information for authentication (optional).	LDAP / Active Directory (Optional)
10	Machine Learning Model	Not applicable for this project.	N/A
11	Infrastructure (Server / Cloud)	Hosts the application and manages workflows securely in the cloud.	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Application Characteristic	Description	Technology Used
1	Front-End	Provides an interactive interface for users to submit and track network requests.	ServiceNow Service Portal, Service Catalog
2	Back-End	Processes requests, manages approvals, updates records, and automates workflows.	ServiceNow Flow Designer
3	Database	Stores request details, assignment information, approval status, and work status.	ServiceNow Custom Tables (u_network_database)
4	Workflow Automation	Automates request creation, assignment, manager approval, and status updates.	ServiceNow Flow Designer
5	Approval Management	Routes requests to the assigned user's manager for approval before execution.	ServiceNow Approval Engine
6	Notification System	Sends email notifications for request submission, approval, rejection, and completion.	ServiceNow Notification Engine
7	Authentication & Security	Provides secure login, role-based access, and data protection.	ServiceNow User Roles, ACLs
8	Cloud Platform	Hosts the application with high availability and scalability.	ServiceNow SaaS Cloud
9	Reporting & Monitoring	Enables tracking of request status and workflow execution.	ServiceNow Reports & Flow Execution Details
10	Integration	Supports integration with email services and enterprise user directories.	SMTP, LDAP / Active Directory

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>